Internship Report

AI-Powered Climate Chatbot Development and Backend Support

Internship at BRIC – National Institute of Plant Genome Research (NIPGR), New Delhi

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Organization: BRIC – National Institute of Plant Genome Research (NIPGR), New Delhi

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1. Introduction

I completed a six-week internship at BRIC–National Institute of Plant Genome Research (NIPGR), where I worked on the **#semanticClimate** project. The project focuses on developing an AI-powered climate chatbot that enables users to ask climate-related questions in natural language and receive reliable, source-based answers using Retrieval-Augmented Generation (RAG).

During the internship, my primary objective was to understand the architecture of the chatbot, contribute to backend-related tasks, perform testing and validation, prepare technical documentation, and develop educational resources for future users.

This internship was my first industry experience and provided practical exposure to real-world AI development, collaborative software engineering, and backend workflows.

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2. Objectives of the Internship

The main objectives of my internship were:

- Understand the architecture of an AI-powered chatbot.
- Learn how Retrieval-Augmented Generation (RAG) works.
- Understand backend workflows and AI pipelines.
- Assist in backend development tasks.
- Perform chatbot testing and validation.
- Create technical documentation and onboarding material.
- Develop educational content explaining the chatbot.
- Explore improvements to the existing chatbot by integrating IPCC reports.

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3. Understanding the Chatbot Architecture

One of the major parts of my internship involved understanding how the chatbot works internally.

I studied the complete backend workflow, including:

- User request handling

- Retrieval pipeline
- Embedding generation
- Semantic search
- ChromaDB retrieval
- Response generation
- Citation generation
- Backend API communication

I created architecture diagrams and workflow explanations to better understand each stage of the pipeline.

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4. Backend Contribution

During the internship, I contributed to backend-related tasks by:

- Understanding backend modules and project structure.
- Assisting in backend workflow implementation.
- Running and testing backend services locally.
- Configuring Cloudflare tunnels.
- Running Redis, Celery, and Ollama.
- Identifying issues during backend execution.
- Exploring improvements to the chatbot pipeline.

This experience helped me understand how backend services communicate to generate responses.

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5. Chatbot Testing and Validation

A significant part of my work involved testing the chatbot.

My responsibilities included:

- Asking different types of climate-related questions.
- Checking response correctness.
- Identifying incorrect retrievals.
- Reporting bugs and unexpected behaviour.
- Suggesting possible improvements.

This helped improve my understanding of chatbot evaluation and AI system validation.

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6. Technical Documentation

I prepared documentation to make the project easier for future contributors.

This included:

- Backend setup instructions.
- Cloudflare configuration.
- Running Redis and Celery.
- Ollama installation.
- Troubleshooting common errors.
- Local project setup guides.

These documents simplify the onboarding process for new developers and interns.

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7. Educational Course Development

One of my major contributions was developing a beginner-friendly course explaining the chatbot.

The course covered:

- Artificial Intelligence
- Large Language Models
- Retrieval-Augmented Generation
- Vector Databases
- ChromaDB
- Semantic Search
- Backend Workflow
- Redis
- Celery
- Ollama
- Flask
- Complete chatbot architecture

The objective was to help high school students and new interns understand the chatbot without requiring advanced programming knowledge.

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8. Technical Presentations

Throughout the internship, I regularly presented my work during weekly meetings.

These presentations included:

- Progress updates

- Backend workflow explanations
- Chatbot architecture
- Testing observations
- Course development progress
- Proposed improvements

Preparing these presentations improved my communication and technical presentation skills.

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9. IPCC Report Integration

Towards the end of the internship, I explored how IPCC reports could be incorporated into the chatbot.

My work included:

- Studying the existing RAG pipeline.
- Understanding document chunking.
- Examining ChromaDB indexing.
- Analysing retrieval behaviour.
- Exploring how IPCC reports could act as an additional knowledge source while preserving the existing Climate Academy retrieval workflow.

Although this work was still in progress, it gave me valuable exposure to extending an existing AI application.

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10. Meetings and Collaboration

The internship followed a collaborative development approach.

I:

- Attended daily online team meetings.
- Participated in technical discussions.
- Presented my weekly progress to the entire team every Monday.
- Received feedback and guidance from mentors.
- Collaborated with other interns on assigned tasks.

These interactions helped me understand how software development teams coordinate work in a professional environment.

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11. Challenges Faced

Some challenges I encountered included:

- Understanding a large existing codebase.

- Learning backend architecture.
- Setting up Redis, Celery, Ollama, and Cloudflare.
- Debugging environment and dependency issues.
- Understanding Retrieval-Augmented Generation (RAG).
- Working with Git and GitHub in a collaborative project.

Each challenge improved my debugging skills, strengthened my problem-solving approach, and increased my confidence in working with complex software systems.